



Java Programming

Case Study 90. Driving School Enrollment & Test Management System

Problem Statement

A driving school needs a system to manage student enrollment, track theory and practical training sessions, and schedule driving tests across two-wheeler and four-wheeler courses. The system should flag students not yet eligible for their final test.

Objectives

- Maintain student and course enrollment information.
- Track theory and practical session attendance.
- Schedule driving tests.
- Determine test eligibility based on completed sessions.
- Search and sort enrollment records.
- Provide a driving school management GUI.

Required Java Implementation

Java Concept	Application
Classes & Objects	Student, Course, TrainingSession, Test
Constructors	Initialize course/session objects
Abstract Class	Course (base for TwoWheelerCourse, FourWheelerCourse)
Inheritance	Course subtype specialization
ArrayList	Store student enrollments
LinkedList	Maintain training session history
HashMap	Student ID → Course details
TreeMap	Students sorted by sessions completed
CRUD	Enrollment and test management
Searching	Search by student/course
Sorting	Sort by sessions completed, test date
Swing	Driving school management GUI
Exception Handling	Invalid session/test data
Validation	Session count, test eligibility, fees



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Expected Modules

1. Student Enrollment
2. Session Scheduling
3. Attendance Tracking
4. Eligibility Determination
5. Test Scheduling
6. Enrollment Search
7. Training Reports

Deliverables

The system uses inheritance to distinguish two-wheeler and four-wheeler training tracks while automating test eligibility checks.
